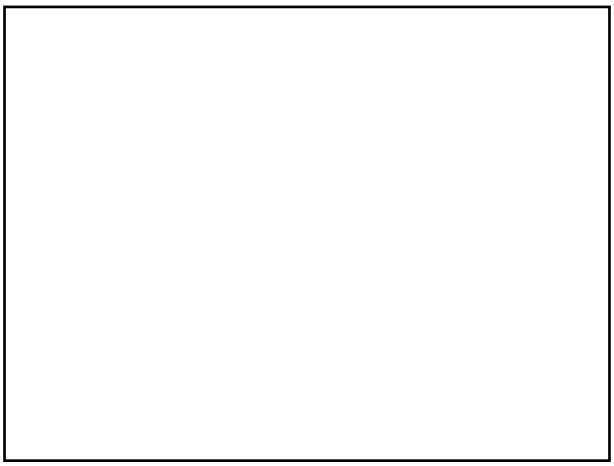ROTATED 90 SYNTHETIC